Week 7 Worksheet — Data Visualization & Outlier Detection

Open-Choice Mini-Project (Sampling Distributions & Correlation Analysis)

Name: _____ **Date:** _____

Project Setup — Choose a Dataset and Tell Its Story
This week you will choose **one dataset among four options** and perform your own statistical exploration. Your goal is to practice:

- Visualizing real datasets using histograms, boxplots, scatter plots, and correlation maps.
- Detecting outliers using the sampling distribution idea (e.g., $\mu \pm k\sigma$).
- Understanding how variables relate to each other (positive, negative, or no correlation).
- Creating at least two visual stories of your choice.

Available Dataset Options (Choose One)

1. **UCI Credit Card Default Dataset** — personal credit limits, bill amounts, payments, and default indicator. Download: `https://archive.ics.uci.edu/dataset/350/default+o`
2. **German Credit (Kaggle)** — loan applicants, demographics, and credit status. Download from Kaggle (account required).
3. **Bankruptcy Prediction — Polish Companies (UCI)** — financial ratios and bankruptcy labels over multiple years.
4. **Yahoo Finance Stocks (yfinance)** — choose any stock (e.g., TSLA, AAPL, SPY). You will download using Python code.

Example: If you choose the UCI Credit Dataset, load the CSV file and map columns to meaningful names (e.g., LIMIT_BAL, EDUCATION, BILL_AMT1).

A. Basic Summary Statistics
Compute basic statistics for at least 6 numeric columns:

- mean, standard deviation, min, max
- number of samples used

Column	Mean	Std	Min	Max

B. Outlier Detection Using Sampling Distribution
Pick your own k for $\mu \pm k\sigma$ (recommend: $k = 2, 2.5, 3$). Report how many rows lie outside the bounds for each selected column.

Column	$\mu - k\sigma$	$\mu + k\sigma$	# Outside	Comment

Hint: Use boxplots or histograms to support your reasoning.

C. Correlation Exploration
Choose at least 4 pairs of variables and compute their correlations. Also include **one visual representation**, such as a scatter plot or heatmap.

Variable Pair	Corr	Interpretation (positive/negative/none)

D. Required Visualizations (Turn in Figures) Produce the following plots:

1. At least **one histogram** (distribution shape).
2. At least **one boxplot** (outliers visible).
3. At least **one scatter plot or correlation heatmap**.
4. **One additional custom visualization** of your choice.

Attach your figures on a separate page or include in your notebook.

E. Reflection (Short Answer) Write 3–5 sentences:

- What did you learn from the dataset you chose?
- Which variable was most surprising?
- Did your outlier rule ($k\sigma$) work well, or would you change it?

Turn-in Checklist

Summary statistics table.

Outlier table using $\mu \pm k\sigma$.

Correlation table and at least one plot.

Four required visualizations.

A short reflection.